

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Data Analysis and Visualization	Course Code:	DL3001
	Program:	BS Data Sciences	Semester:	Fall 2023
	Section	BS DS 5A, 5D	Total Marks:	50
	Due Date:	20 th October 2023	Weightage:	
	Exam Type:	Assignment 2	Page(s):	2

Instructions:

- Submit the solution in a zipped file named as your roll number., i.e., 21L-1234.zip
- You are not allowed to copy solutions from other students (same/cross section). Your code will be checked for plagiarism. If any sort of cheating is found, heavy penalties will be given to all students involved.
- Late submission of your solution is not allowed.

Question 1: Image Processing [25 Marks]

You are provided with the image dataset of cats and dogs. You have to write Python code and implement the following tasks:

- Convert RGB images to grayscale using following formula:
$$\text{grayscale image} = ((0.3 * R) + (0.59 * G) + (0.11 * B))$$
- Design a 3x3 convolution and correlation filter for this dataset.
- Design 1st order derivative and 2nd Order derivative filter and apply on input images.
- Identify the type of noise (Gaussian or Salt and Pepper) in images.
- Based on your assumption apply relative filter on input images to remove the noise.
- Implement Laplacian edge detection on your dataset.

Question 2: Text Processing [25 Marks]

You are provided with a labelled dataset that contains data of tweets with their corresponding labels (0 for hate speech and 1 for neutral text). Use the given dataset to perform the following tasks:

- Load the dataset using pandas and split it into training and testing sets.
- Implement the Bag of Words Feature Extraction Technique.
- Implement the TF-IDF Feature Extraction Technique.

- Preprocess the provided data with **inbuilt libraries** using:
 - Tokenization: Tokenize the text into individual words or tokens.
 - Lowercasing: Convert all tokens to lowercase.
 - Stop word Removal: Remove common English stop words, i.e., a, is, am, the, as, of, to, from.
 - Smoothing: Implement Laplace smoothing to handle out-of-vocabulary words.
 - Normalization: Normalize the text by removing punctuation and special characters.
 - Stemming: Apply stemming to reduce words to their base form.
 - Lemmatization: Implement lemmatization to reduce words to their dictionary form.
- Build a logistic regression model from scratch (**do not use inbuilt function**) for the text classification. Train the logistic regression model using both the BOW and TF-IDF representations. (Use **sigmoid activation function**).
- Evaluate the performance of the logistic regression model using the accuracy, precision, recall, and F1-score for both classes.
- Compare the performance of the logistic regression model trained on BOW and TF-IDF representations.

Happy Coding!